

UNIT – 5: GOVERNANCE, REGULATION, AND CASE STUDIES

Case Studies

1. Microsoft Tay (2016)

Overview: Tay was a Twitter-based chatbot created by Microsoft to learn conversational language from real users. It was designed as an experiment to mimic teenage speech patterns.

What Went Wrong: Within hours of release, Tay began posting **racist, abusive, and offensive** tweets. Users deliberately sent toxic messages, and Tay rapidly copied them.

Why It Happened

- **Online learning without filters:** Tay learned directly from live user input with no safety controls.
- **No toxicity or content moderation system:** The model lacked filters to block harmful content.
- **Poor adversarial testing:** Microsoft did not anticipate coordinated attacks from malicious users.

Mitigation & Lessons

- Stop real-time learning from untrusted data.
 - Use strong **content filters** and **toxicity detectors**.
 - Test the system against adversarial prompts before launch.
 - Gradual or controlled rollout, not open public release.
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2. Meta Galactica (2022)

Overview: Galactica was an LLM designed to assist with scientific tasks by generating summaries, papers, and scientific explanations.

What Went Wrong: Galactica produced **confident but incorrect scientific facts, fake citations, and fabricated research** — leading to widespread criticism. Meta removed the model shortly after launch.

Why It Happened

- **Hallucinations:** The model created information that sounded correct but was false.
- **Unverified training data:** Mixed-quality scientific sources caused unreliable outputs.
- **Lack of fact-checking or retrieval mechanisms:** It had no way to verify claims against trusted databases.

Mitigation & Lessons

- Use **retrieval-based grounding** for scientific or factual systems.

- Provide **verifiable citations** instead of invented ones.
 - Deploy to researchers only after strict evaluation.
 - Include disclaimers, safeguards, and limited access.
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3. OpenAI ChatGPT (2022–present)

Overview: ChatGPT is a general-purpose conversational AI widely used globally. It is the most influential LLM deployment to date.

Common Issues Observed

- **Hallucinations (incorrect answers)**
- **Biases and stereotypes** inherited from training data
- **Privacy concerns** raised by regulators (e.g., Italy’s investigation)
- **Jailbreaking and prompt injection** attempts that bypass safety rules

Why These Issues Occur

- **Training on large internet datasets** containing bias and misinformation
- **LLM objective favors fluency, not accuracy**
- **Safety filters are imperfect** and adversarial prompts can bypass them

Mitigation & Improvements

- **RLHF (Reinforcement Learning from Human Feedback)** for safer outputs
 - **Safety classifiers and output filters**
 - **Rate limits and content moderation pipelines**
 - **Transparency and compliance** with GDPR and other laws
 - **Continuous red-teaming and updates** as threats evolve
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Analysis of Mitigation Strategies and Lessons Learned

Key Mitigation Strategies

A. Data Governance and Bias Reduction

AI behaviour heavily depends on the quality of its training data.

Strong data governance helps prevent biased outputs, hallucinations, or harmful recommendations.

Important practices include:

- **Dataset auditing:** checking for bias, imbalance, or inappropriate content
- **Data cleaning:** removing offensive, misleading, or low-quality samples
- **Documentation:** maintaining datasheets, model cards, and sources
- **Bias mitigation techniques:** rebalancing, sampling, synthetic data

Impact:

- Improves fairness and accuracy
- Reduces discriminatory outputs

Limitation: Bias cannot be completely eliminated because datasets reflect real-world inequalities.

B. Model-Level Safety Mechanisms

1. Reinforcement Learning from Human Feedback (RLHF): Human evaluators rate model responses. The model learns to prefer safe, helpful, and policy-aligned outputs.

Benefit:

- Greatly reduces toxic or harmful responses
- Makes the model more polite, compliant, and predictable

2. Safety Fine-Tuning: After the base model is trained, it is retrained using curated "safe" datasets.

Benefit:

- Strengthens refusal behaviour (e.g., rejecting illegal requests)
- Helps align the model with ethical guidelines

3. Adversarial Training: The model is trained using intentionally harmful or confusing prompts.

Benefit:

- Improves model resilience against prompt injection, jailbreaking, and manipulation
 - Helps developers identify weak spots before deployment
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C. Content Filtering and Moderation Systems: Even a safe model can produce unsafe outputs. Therefore, an additional **filtering layer** is used to monitor both input and output.

Techniques include:

- Keyword and pattern matching
- Toxicity, hate-speech, and spam classifiers
- Allow-list responses for restricted domains
- Output structure constraints (e.g., JSON format)
- Human-in-the-loop escalation for critical cases

Strength: Acts as a “last defence layer” even if the model misbehaves.

Weakness: Clever attackers can bypass filters using unusual characters or coded language.

D. Human Oversight and Review: No AI system is fully reliable. Human oversight ensures that high-impact decisions are reviewed by humans.

Forms of oversight:

- Approval before executing sensitive actions
- Review of flagged model outputs
- Expert supervision in high-risk domains (healthcare, law, finance)

Importance: Prevents AI from taking harmful or irreversible actions.

E. Transparency, Logging, and Monitoring: Transparency increases trust and helps regulators assess how AI systems behave.

Key practices:

- Inform users when interacting with AI
- Document model limitations and intended uses
- Maintain logs of model decisions and system interactions
- Conduct periodic audits

Benefit: Ensures accountability and easier investigation of harmful outputs.

F. Regulatory and Policy Frameworks: Governments and organizations provide rules to ensure responsible AI deployment.

Examples:

- **EU AI Act** (risk-based regulation)
- **GDPR** (data privacy)
- **Organizational policies:** internal audits, ethical guidelines, safety testing

Importance: Provides legal structure and compliance requirements to prevent harmful AI deployment.

Lessons Learned from Past Failures

Lesson 1: AI Must Not Learn Directly from Unfiltered User Input

Tay learned in real time from Twitter users.

Result → model started producing hate speech within hours.

Takeaway: AI training pipelines must be controlled, filtered, and monitored.

Lesson 2: Accuracy Alone Is Not Enough

Galactica generated scientific-sounding but incorrect information.

Takeaway:

In high-risk fields, AI must also be:

- Evidence-based
- Transparent
- Verifiable
- Grounded in reliable sources

Lesson 3: Guardrails Must Be Strong and Continuously Updated

Early ChatGPT versions were easily jailbroken.

Takeaway: Safety training must be ongoing because new attack methods appear constantly.

Lesson 4: Human Oversight Is Essential in High-Stakes Domains

Fully automated decisions can be dangerous.

Takeaway: AI should support, not replace, human decision-makers.

Lesson 5: Monitoring Continues After Deployment

AI behaviour can change due to:

- New user interactions
- New attack patterns
- Updated data

Takeaway: AI safety is a continuous process, not a one-time task.

Lesson 6: Transparency Improves Trust and Accountability

Users must understand when AI is used and what limitations it has.

Takeaway: Clear policies and disclosures reduce misuse and increase public confidence.